1 Differentiability

We will eventually want to use derivatives in the multivariable setting for the same sort of applications that we solved using the derivative in single-variable calculus (e.g., constrained optimization problems, etc.). To do so, we will need to use many of the same theoretical properties of the derivative.

For example, recall from one-variable calculus that differentiability is a stronger condition than continuity. If f is differentiable at a then

$$\begin{aligned}\lim_{x\to a} f(x) &= \lim_{x\to a} \frac{f(x)-f(a)}{x-a}(x-a)+f(a)\\ &= \lim_{x\to a} \frac{f(x)-f(a)}{x-a} \lim_{x\to a}(x-a)+\lim_{x\to a} f(a)\\ &= f'(a)\cdot 0+f(a)\\ &= f(a),\end{aligned}$$

so f is continuous at a. However, $f(x)=|x|$ is continuous at 0, but not differentiable at 0.

Suppose now that $f(x,y)$ is a function of two variables whose partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ both exist at (a,b). Does it follow that f is necessarily continuous at (a,b)? It turns out that the answer is no.

Example 1.1. Let $f:\mathbb{R}^2\to\mathbb{R}$ be defined by

$$f(x,y)=\begin{cases}\frac{x^2y^2}{x^4+y^4} & \text{if } (x,y)\neq(0,0),\\ 0 & \text{if } (x,y)=(0,0).\end{cases}$$

The partial derivatives at the origin both exist, since

$$\begin{aligned}\frac{\partial f}{\partial x}(0,0) &= \lim_{h\to 0}\frac{f(0+h,0)-f(0,0)}{h} = \lim_{h\to 0}\frac{0}{h} = \lim_{h\to 0} 0 = 0\\ \frac{\partial f}{\partial y}(0,0) &= \lim_{h\to 0}\frac{f(0,0+h)-f(0,0)}{h} = \lim_{h\to 0}\frac{0}{h} = \lim_{h\to 0} 0 = 0.\end{aligned}$$

However, $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist, since approaching the origin along the x-axis gives

$$\lim_{x\to 0}\frac{0}{x^4} = \lim_{x\to 0} 0 = 0$$

while approaching along the line $y=x$ gives

$$\lim_{x\to 0}\frac{x^4}{2x^4} = \lim_{x\to 0}\frac{1}{2} = \frac{1}{2}.$$

Continuity at (a,b) requires that $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$ when the limit is taken along *any* path passing through (a,b). The partial derivatives, on the other hand, only depend on what is happening along lines through (a,b) parallel to the x and y axes, and therefore they failed to detect the discontinuity in the previous example. This suggests we need a stronger condition

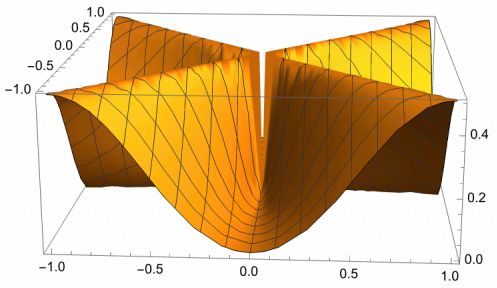


Figure 1: The graph of the function in Example 1.1.

in our definition of differentiability for multivariable functions than the existence of the partial derivatives if we want differentiable multivariable functions to behave similarly to differentiable single variable functions.

To this end, let's rewire the definition of the derivative of a single-variable function $f(x)$ in a way that will generalize to multivariable functions. If $f'(a)$ exists, then

$$\begin{aligned}
\lim_{x\to a}\frac{f(x)-f(a)}{x-a} &= f'(a)\\
\lim_{x\to a}\frac{f(x)-f(a)}{x-a} &= \lim_{x\to a} f'(a)\\
\lim_{x\to a}\frac{f(x)-f(a)}{x-a} - \lim_{x\to a} f'(a) &= 0\\
\lim_{x\to a}\left(\frac{f(x)-f(a)}{x-a} - f'(a)\right) &= 0\\
\lim_{x\to a}\left(\frac{f(x)-f(a)}{x-a} - \frac{f'(a)(x-a)}{x-a}\right) &= 0\\
\lim_{x\to a}\left(\frac{f(x)-f(a)-f'(a)(x-a)}{x-a}\right) &= 0\\
\lim_{x\to a}\left(\frac{f(x)-(f(a)+f'(a)(x-a))}{x-a}\right) &= 0
\end{aligned}$$

This shows that, if f is differentiable at a, then the difference between f and the function h defined by $h(x) = f(a) + f'(a)(x-a)$ goes to zero faster than $x-a$ goes to zero. The graph $y = h(x)$ of the function h is of course just the tangent line to the graph of f at $(a, f(a))$. This is what is meant when one says that the tangent line at $(a, f(a))$ is the "best linear approximation" to the graph of f near $(a, f(a))$.

We take this idea of having a "best linear approximation" as the defining property of a differentiability at a point. The best linear approximation to f at a is called the *differential of f at a*. As differentials may not have been emphasized in your Calculus I course, we now explain this in some detail.

2 The Differential

Definition 2.1 (Linear, affine function)**.** (a) A function $f:\mathbb{R}^n\to\mathbb{R}$ is *linear* if $f(\mathbf{x})=A\mathbf{x}$ for some $1\times n$ constant matrix A.

(b) A function $f:\mathbb{R}^n\to\mathbb{R}^m$ is *affine* if $f(\mathbf{x})=A\mathbf{x}+b$ for some $1\times n$ constant matrix A and constant $b\in\mathbb{R}$.

Note that every linear function is affine (with $b=0$), but an affine function with $b\neq 0$ is not linear. Given an affine function $f(\mathbf{x})=A\mathbf{x}+b$, the associated linear function $L(\mathbf{x})=A\mathbf{x}$ is called the *differential* of f, which we denote by $L=df$.

Example 2.2. (a) An affine function $f:\mathbb{R}\to\mathbb{R}$ is of the form $f(x)=ax+b$ for some constants $a,b\in\mathbb{R}$. Its differential is $df(x)=ax$.

(b) An affine function $f:\mathbb{R}^2\to\mathbb{R}$ is of the form $f(x,y)=ax+by+c$ for some constants $a,b,c\in\mathbb{R}$. Its differential is $df(x)=ax+by$.

(c) An affine function $f:\mathbb{R}^3\to\mathbb{R}$ is of the form $f(x,y,z)=ax+by+cz+d$ for some constants $a,b,c,d\in\mathbb{R}$. Its differential is $df(x,y,z)=ax+by+cz$.

We have seen above that if a single-variable function f is differentiable at a, then the difference between $f(x)$ and the affine function $h(x)=f(a)+f'(a)(x-a)$ goes to zero faster than $x-a$ goes to zero. Denoting $\Delta y=f(x)-f(a)$ and $\Delta x=x-a$, this can be equivalently stated by saying that the difference between Δy (which is a function of x) and the linear function $f'(a)\Delta x$ (which is also a function of x) goes to zero faster than Δx goes to zero.

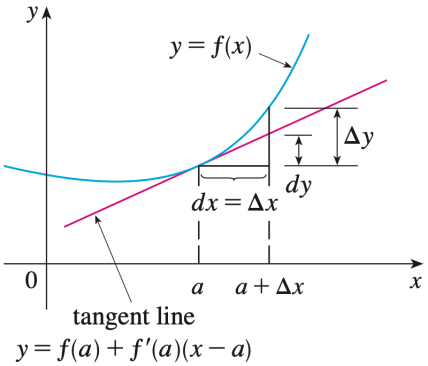


Figure 2: If f is differentiable at a, then $\Delta y-dy$ goes to zero faster than Δx.

We define the *differential of f at a*, denoted by df_a, by $df_a=dh$. That is, $df_a:\mathbb{R}\to\mathbb{R}$ is the *linear* function

$$df_a(x)=f'(a)x,$$

so that $\Delta y\approx df_a(\Delta x)=f'(a)\Delta x$ for Δx small. Geometrically, Δy is the change in the graph of f as $a\mapsto a+\Delta x$ while $df_a(\Delta x)$ is the change in the tangent line as $a\mapsto a+\Delta x$. Since the notation

$df_a(\Delta x)$ is rather cumbersome, textbooks usually write the input Δx as dx and the output $df_a(dx)$ as dy, so that $dy = f'(a)dx$. Writing $f'(a) = \frac{dy}{dx}$, this becomes

$$dy = \frac{dy}{dx}dx.$$

The *linear approximation* [1] (near $x = a$) of a function f differentiable at a is the function

$$h(x) = f(a) + f'(a)(x-a).$$

Example 2.3. The differential of the function $y = x^2$ at any point x is $dy = 2xdx$. The change in the function if we change $x = 2$ to $x + dx = 2.01$ is approximately

$$\Delta y \approx dy = 2(2)(2.01-2) = 4(.01) = .04.$$

The value of the function at 2.01 is therefore approximately

$$y = f(2.01) \approx f(2) + dy = 4 + .04 = 4.04.$$

The exact value of the function at 2.01 is $(2.01)^2 = 4.0401$, so the relative error in this approximation is $\frac{.0001}{4} = 2.5 \times 10^{-5}$ or 0.0025%.

The notion of the differential of a function makes sense for multivariable functions. If f is a function of two variables, then the differential of f at (a,b) is a linear function $df_{(a,b)}(x,y) = Ax + By$ so that the difference between $\Delta z = f(x,y) - f(a,b)$ and $df_{(a,b)}(\Delta x, \Delta y)$ goes to zero faster than $||(x,y) - (a,b)||$ goes to zero; that is

$$\lim_{(x,y)\to(a,b)} \frac{f(x,y) - f(a,b) - A(x-a) - B(y-b)}{||(x,y) - (a,b)||} = 0. \tag{2.4}$$

If this limit exists, then it must exist along any path from (x,y) to (a,b). In particular, along the straight-line path $y = b$, it becomes

$$\lim_{x\to a} \frac{f(x,b) - f(a,b) - A(x-a)}{|x-a|} = 0. \tag{2.5}$$

Since $|x-a| = \pm(x-a)$, this implies

$$\begin{aligned} 0 &= \lim_{x\to a} \frac{f(x,b) - f(a,b) - A(x-a)}{x-a} \\ &= \lim_{x\to a} \left(\frac{f(x,b) - f(a,b)}{x-a} - A \right) \\ &= \lim_{x\to a} \frac{f(x,b) - f(a,b)}{x-a} - \lim_{x\to a} A \\ &= \lim_{x\to a} \frac{f(x,b) - f(a,b)}{x-a} - A, \end{aligned}$$

[1] 'Affine approximation' would be a better name, but this is what it is called in most textbooks.

so the existence of the limit in Equation 2.4 implies that the partial derivative with respect to x exists, and that $\frac{\partial f}{\partial x}(a,b) = A$. Similarly, taking the limit in Equation 2.4 along the straight line path $x = a$ shows that the partial derivative with respect to y exists, and that $\frac{\partial f}{\partial y}(a,b) = B$. This shows that if the differential of $f(x,y)$ exists at (a,b), then it is unique and is given by

$$df_{(a,b)}(\Delta x) = \frac{\partial f}{\partial x}(a,b)(x-a) + \frac{\partial f}{\partial y}(a,b)(y-b).$$

The graph $z = h(x,y)$ of the corresponding affine function

$$h(x,y) = f(a,b) + \frac{\partial f}{\partial x}(a,b)(x-a) + \frac{\partial f}{\partial y}(a,b)(y-b)$$

is a plane, called the *tangent plane* to the graph of f at (a,b).

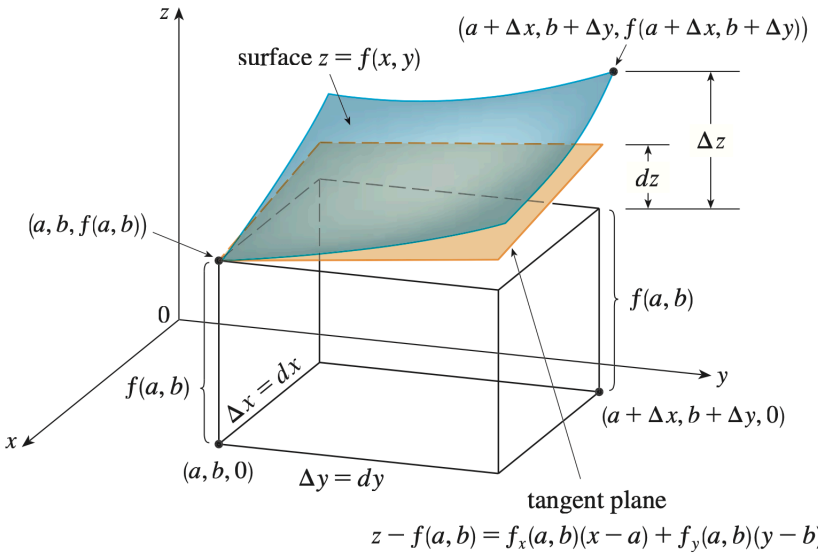


Figure 3: The tangent plane to the graph of a differentiable function.

As before, we write $dx = \Delta x = x - a$, $dy = \Delta y = y - b$, and $dz = df_{(a,b)}(dx, dy)$, so that

$$dz = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy.$$

Example 2.6. The differential of $z = x^2 + y^2$ at any point (x,y) is given by $dz = 2xdx + 2ydy$. Then change in the function if we change $(x,y) = (2,3)$ to $(x + dx, y + dy) = (2.01, 3.02)$ is approximately

$$\Delta z \approx 2(2)(2.01 - 2) + 2(3)(3.02 - 3) = 4(.01) + 6(.02) = .04 + .12 = .16.$$

The value of the function at $(2.01, 3.02)$ is therefore approximately $f(2.01, 3.02) \approx f(2,3) + dz = 13.16$. The exact value of the function is $f(2.01, 3.02) = (2.01)^2 + (3.02)^2 = 13.1605$. The relative error in this approximation is therefore $\frac{.0005}{13} \approx 3.8 \times 10^{-4}$ or $.038\%$.

Definition 2.7 (Differentiable function)**.** We say that a multivariable function $z = f(x,y)$ is *differentiable* at (a,b) if

$$\lim_{(x,y)\to(a,b)} \frac{\Delta z - dz}{||(x,y)-(a,b)||} = 0.$$

where $dz = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$. We define differentiability in the same way for a function of $n > 2$ variables.

If f is differentiable at (a,b), then the *linear approximation* of f (near (a,b)) is the function

$$h(x,y) = f(a,b) + \frac{\partial f}{\partial x}(a,b)(x-a) + \frac{\partial f}{\partial y}(a,b)(y-b).$$

Example 2.8. If $f(x,y) = c$ is a constant function, then $\Delta z = f(x,y) - f(a,b) = c - c = 0$ and

$$\begin{aligned} dz &= \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy \\ &= 0dx + 0dy \\ &= 0, \end{aligned}$$

and

$$\lim_{(x,y)\to(a,b)} \frac{\Delta z - dz}{||(x,y)-(a,b)||} = \lim_{(x,y)\to(a,b)} 0 = 0,$$

which shows that a constant function is differentiable everywhere with its differential at every point being the zero map (the linear function which maps all of $\mathbb{R}^2$ to 0).

Example 2.9. Consider the function $f(x,y) = Ax + By + C$. Then

$$\begin{aligned} \Delta z &= A(x-a) + B(y-a) \\ &= Adx + Bdy \end{aligned}$$

and

$$\begin{aligned} dz &= \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy \\ &= Adx + Bdy \end{aligned}$$

and therefore

$$\lim_{(x,y)\to(a,b)} \frac{\Delta z - dz}{||(x,y)-(a,b)||} = \lim_{(x,y)\to(a,b)} \frac{0}{||(x,y)-(a,b)||} = \lim_{(x,y)\to(a,b)} 0 = 0,$$

so an affine function is differentiable everywhere and its differential is equal to its differential as an affine function.

3 Differentiability implies continuity

We now prove that, with the definition of differentiability above, differentiability at $\mathbf{a}$ implies continuity at $\mathbf{a}$. It will be convenient to write $\mathbf{a} = (a,b)$, $\mathbf{h} = (x-a, y-b)$, $df_{(a,b)}(\Delta x, \Delta y) = df_{\mathbf{a}}(\mathbf{h})$ so that the limit in the definition of differentiability at $\mathbf{a}$ is equivalent to

$$\lim_{\mathbf{h}\to\mathbf{0}} \frac{f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - df_{\mathbf{a}}(\mathbf{h})}{||\mathbf{h}||} = 0.$$

More precisely, this says that, given any $\varepsilon > 0$ there exists $\delta > 0$ such that $||\mathbf{h}|| < \delta$ implies $\frac{|f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - df_{\mathbf{a}}|}{||\mathbf{h}||} < \varepsilon$ or, equivalently, $|f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - df_{\mathbf{a}}| < \varepsilon||\mathbf{h}||$.

Theorem 3.1. *If f is differentiable at $\mathbf{a}$, then f is continuous at $\mathbf{a}$.*

We will need the following Lemma.

Lemma 3.2. *If $f(\mathbf{x}) = A\mathbf{x}$ is linear, then there exists a real number $K \geq 0$ such that $||f(\mathbf{x})|| \leq K||\mathbf{x}||$ for all $\mathbf{x} \in \mathbb{R}^n$.*

Proof. (of Lemma) Write $A\mathbf{x} = A_{11}x + A_{12}y$ and let $M = \max\{A_{11}, A_{12}\}$. Then

$$\begin{aligned} ||A\mathbf{x}||^2 &= A_{11}^2x^2 + A_{12}^2y^2 \\ &\leq M^2x^2 + M^2y^2 \\ &= M^2(x^2+y^2). \end{aligned}$$

Since both sides are nonnegative, this implies

$$||A\mathbf{x}|| \leq \sqrt{M^2}\sqrt{x^2+y^2} = |M|||\mathbf{x}||.$$

This shows that $||A\mathbf{x}|| \leq K||\mathbf{x}||$ for all $\mathbf{x} \in \mathbb{R}^n$, where $K = |M| = |\max\{A_{11}, A_{12}\}| \geq 0$. The proof is essentially the same for $n > 2$ variables. □

Proof. (of Theorem) Suppose f is differentiable at $\mathbf{a}$. We need to show that $\lim_{\mathbf{h}\to\mathbf{0}} f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a})$. Let $\varepsilon > 0$. Then

$$\begin{aligned} |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})| &= |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - df_{\mathbf{a}}(\mathbf{h}) + df_{\mathbf{a}}(\mathbf{h})| \\ &\leq |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - df_{\mathbf{a}}(\mathbf{h})| + |df_{\mathbf{a}}(\mathbf{h})| \quad \text{(by the Triangle Inequality)} \end{aligned}$$

Since f is differentiable at $\mathbf{a}$, we can find $\delta_0 > 0$ such that $||\mathbf{h}|| < \delta_0$ implies

$$|f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - df_{\mathbf{a}}(\mathbf{h})| < ||\mathbf{h}||.$$

Since $df_{\mathbf{a}}$ is linear, by the preceding lemma there exists $K \geq 0$ such that $|df_{\mathbf{a}}(\mathbf{h})| \leq K||\mathbf{h}||$. Thus, if $||\mathbf{h}|| < \delta_0$, then

$$\begin{aligned} |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})| &\leq |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - df_{\mathbf{a}}(\mathbf{h})| + |df_{\mathbf{a}}(\mathbf{h})| \\ &< ||\mathbf{h}|| + K||\mathbf{h}|| \\ &= (1+K)||\mathbf{h}||. \end{aligned}$$

It follows that, by taking $||\mathbf{h}|| < \delta = \min\{\delta_0, \frac{\varepsilon}{1+K}\}$, we have

$$\begin{aligned} |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})| &< (1+K)||\mathbf{h}|| \\ &< (1+K)\frac{\varepsilon}{1+K} \\ &= \varepsilon, \end{aligned}$$

which proves that $\lim_{\mathbf{h}\to\mathbf{0}} f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a})$. □

The next theorem gives a convenient way to check if a function is differentiable at a point **a**.

Theorem 3.3. *If the partial derivatives of $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ exist and are continuous throughout some open set containing* **a***, then f is differentiable at a.*

Proof. Suppose the partial derivatives of $f(x,y)$ exist at (a,b) an are continuous on some open set containing (a,b). To show that f is differentiable at (a,b), we need to show that

$$\lim_{(x,y)\to(a,b)} \frac{f(x,y) - f(a,b) - \frac{\partial f}{\partial x}(a,b)(x-a) - \frac{\partial f}{\partial y}(a,b)(y-b)}{||(x-y)-(a,b)||} = 0.$$

First, note that

$$f(x,y) - f(a,b) = f(x,y) - f(a,y) + f(a,y) - f(a,b). \tag{3.4}$$

Since $\frac{\partial f}{\partial x}$ exists throughout an open set containing (a,b), for (x,y) close enough to (a,b) the Mean Value Theorem (from one-variable calculus) tells us there exists c_1 between a and x such that

$$f(x,y) - f(a,y) = \frac{\partial f}{\partial x}(c_1,y)(x-a).$$

Similarly, there exists c_2 between b and y such that

$$f(x,y) - f(x,b) = \frac{\partial f}{\partial y}(x,c_2)(y-b).$$

Therefore

$$\begin{aligned} &f(x,y) - f(a,b) - \frac{\partial f}{\partial x}(a,b)(x-a) - \frac{\partial f}{\partial y}(a,b)(y-b) \\ &= [\frac{\partial f}{\partial x}(c_1,y) - \frac{\partial f}{\partial x}(a,b)](x-a) + [\frac{\partial f}{\partial y}(x,c_2) - \frac{\partial f}{\partial y}(a,b)](y-b) \\ &\le |[\frac{\partial f}{\partial x}(c_1,y) - \frac{\partial f}{\partial x}(a,b)](x-a) + [\frac{\partial f}{\partial y}(x,c_2) - \frac{\partial f}{\partial y}(a,b)](y-b)| \\ &\le |\frac{\partial f}{\partial x}(c_1,y) - \frac{\partial f}{\partial x}(a,b)||x-a| + |\frac{\partial f}{\partial y}(x,c_2) - \frac{\partial f}{\partial y}(a,b)||y-b| \quad \text{(by the Triangle Inequality)} \\ &\le |\frac{\partial f}{\partial x}(c_1,y) - \frac{\partial f}{\partial x}(a,b)|||\mathbf{x}-\mathbf{a}|| + |\frac{\partial f}{\partial y}(x,c_2) - \frac{\partial f}{\partial y}(a,b)|||\mathbf{x}-\mathbf{a}|| \\ &= \left(\left|\frac{\partial f}{\partial x}(c_1,y) - \frac{\partial f}{\partial x}(a,b)\right| + \left|\frac{\partial f}{\partial y}(x,c_2) - \frac{\partial f}{\partial y}(a,b)\right|\right)||\mathbf{x}-\mathbf{a}|| \end{aligned}$$

and it follows that

$$\frac{f(x,y)-f(a,b)-\frac{\partial f}{\partial x}(a,b)(x-a)-\frac{\partial f}{\partial y}(a,b)(y-b)}{||\mathbf{x}-\mathbf{a}||} \leq \left|\frac{\partial f}{\partial x}(c_1,y)-\frac{\partial f}{\partial x}(a,b)\right|+\left|\frac{\partial f}{\partial y}(x,c_2)-\frac{\partial f}{\partial y}(a,b)\right|$$

Since the partial derivatives of f are continuous on an open set containing $\mathbf{a}$, we have

$$\lim_{(x,y)\to(a,b)}\left(\frac{\partial f}{\partial x}(x,y)-\frac{\partial f}{\partial x}(a,b)\right)=0$$

and

$$\lim_{(x,y)\to(a,b)}\left(\frac{\partial f}{\partial y}(x,y)-\frac{\partial f}{\partial x}(a,b)\right)=0.$$

Since c_1 is between x and a, $c_1 \to a$ as $x \to a$. Similarly, since c_2 is between y and b, $c_2 \to b$ as $y \to b$. Thus, both terms on the right hand side go to zero as $(x,y) \to (a,b)$, proving that f is differentiable at (a,b). The proof for a function of $n > 2$ variables is essentially the same. □

Example 3.5. Let $f(x,y) = xy - 7x^8y^2 + \cos(x)$. Since

$$\frac{\partial f}{\partial x} = y - 56x^7y^2 - \sin(x)$$

and

$$\frac{\partial f}{\partial y} = x - 14x^8y$$

are continuous everywhere, f is differentiable everywhere.

Example 3.6. Recall the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} \frac{x^2y^2}{x^4+y^4} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

from Example 1.1. If $(x,y) \neq (0,0)$, then

$$\frac{\partial f}{\partial x} = -\frac{2xy^2(x^4-y^4)}{(x^4+y^4)^2}$$

and

$$\frac{\partial f}{\partial y} = \frac{2x^2y(x^4-y^4)}{(x^4+y^4)^2}$$

which are continuous everywhere except $(0,0)$. We have already seen that f is discontinuous at $(0,0)$, so f can't be differentiable there. Since the partial derivatives are continuous on the open set $\mathbb{R}^2 - \{(0,0)\}$, the function f is differentiable everywhere except the origin.

Name:_____

4 Summary

- A function $f(x,y)$ is defined to be differentiable at (a,b) if it has a tangent plane at (a,b). The equation of the tangent plane is

$$z = f(a,b) + \frac{\partial f}{\partial x}(a,b)(x-a) + \frac{\partial f}{\partial y}(a,b)(y-b).$$

- If f is differentiable at (x,y), and if we change x and y by small amounts dx and dy, respectively, then $\Delta z = f(x+dx, y+dy) - f(x,y) \approx dz$, where dz is the *differential of the function f at (x,y)*, defined by

$$dz = \frac{\partial f}{\partial x}(x,y)dx + \frac{\partial f}{\partial y}(x,y)dy.$$

- If f is differentiable at (x,y), then both partial derivatives exist at (x,y), but existence of both partial derivatives at (x,y) is not sufficient go guarantee that f is differentiable at (x,y).

- However, if both partial derivatives exist and are *continuous* on an open set containing (x,y), then f is differentiable at (x,y).